Where to Practise? Regret-Based Pole Sampling and Stability Shaping for a Generalist DQN CartPole Agent

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ABSTRACT

We train a fixed 4–128–128–2 Double DQN to balance CartPole for pole lengths 0.4–1.8 and test it under unseen wind and force perturbations. We compare uniform domain randomisation, regret-based adaptive pole sampling, and adaptive sampling with a stability-shaped reward (3 seeds each). The shaped agent scores 10,065 steps on average versus 6,531 for uniform and 2,246 for adaptive sampling: rewarding *how well* the pole is balanced mattered more than choosing *which* lengths to practise on.

KEYWORDS

deep Q-learning, generalisation, curriculum learning, reward shaping, CartPole

1 INTRODUCTION

CartPole [1] is a classic control benchmark, but a policy trained for a single pole length rarely transfers to other lengths: a short pole falls fast and needs quick corrections, while a long pole is slow but punishes over-correction. In this work a single Deep Q-Network (DQN) [4] with a fixed architecture (4–128–128–2) must act as a *generalist* over pole lengths $\ell \in [0.4, 1.8]$, and is tested under conditions it never sees during training: wind bumps after 500 steps and a growing force magnitude after 1000 steps.

The standard answer to generalisation over environment parameters is domain randomisation [7]: sample the parameter uniformly during training. Uniform sampling, however, spends equal effort on lengths the agent has already mastered and on lengths it still fails. Curriculum learning [2] and Prioritized Level Replay [3] suggest that the training distribution should instead follow the learner's weaknesses. Separately, surviving the unseen perturbations requires a *safety margin*, which the plain +1-per-step reward does not encourage: any policy that keeps the pole up for 500 steps is optimal.

Research question. *How do (i) the way pole lengths are sampled during training and (ii) rewarding stability rather than mere survival affect the robustness of a DQN generalist to unseen pole lengths and perturbations?* We compare three strategies: uniform domain randomisation (baseline), regret-based adaptive pole sampling, and adaptive sampling combined with a stability-shaped reward. Besides the average test score we look at *where* over the length range each strategy succeeds or fails and at the variance across training seeds.

2 METHODS

2.1 Common DQN learner

All strategies share the same learner, so that the only difference is the sampling of ℓ and the reward. We use DQN with experience replay [4], the Double-DQN target [8]

$$y = r + \gamma(1-d)\, Q_{\bar{\theta}}\big(s', \arg\max_a Q_\theta(s', a)\big),$$

a Polyak-averaged target network ($\tau = 0.005$), Huber loss and gradient-norm clipping at 10. Hyperparameters: Adam, learning rate $5 \cdot 10^{-4}$, $\gamma = 0.99$, batch 128, replay buffer 10^5 transitions, 2,000 warm-up steps, one gradient step per environment step, and ϵ-greedy exploration decayed linearly from 1.0 to 0.02 over the first 24,000 steps. These are standard CartPole DQN settings; the soft target update and Double-DQN reduce the Q-value overestimation and instability that otherwise make CartPole DQN collapse late in training. The 500-step time-limit is treated as truncation (bootstrapped), not as a terminal state. A new ℓ is drawn at every episode; only `env.unwrapped.length` is changed, all other dynamics are default.

Model selection. Every 5,000 steps the greedy policy is validated on five held-out lengths $\{0.4, 0.75, 1.1, 1.45, 1.8\}$ (2 episodes each, default dynamics, 500 steps). We keep the checkpoint with the longest mean survival, breaking ties by the lowest mean $|\theta|$. The test script is never used for training or selection.

2.2 Strategy 1: Uniform domain randomisation (baseline)

$\ell \sim \mathcal{U}(0.4, 1.8)$ each episode [7], reward +1 per step. This is the natural baseline: every length is seen equally often.

2.3 Strategy 2: Regret-based adaptive pole sampling

Inspired by Prioritized Level Replay [3], the range is split into $K = 15$ equal bins. Each bin k keeps a competence score c_k, an exponential moving average ($\alpha = 0.1$) of the normalised episode length $T/500$ of training episodes in that bin (initially 0). A bin is sampled with probability $p_k \propto (1 - c_k + 0.05)^{1/0.5}$ and ℓ uniformly within it. Lengths the agent currently fails get sampled more ("practise where you are weak"), the floor 0.05 keeps every bin in the distribution to avoid forgetting, and the temperature 0.5 sharpens the preference. The motivation is that the hard lengths (very short poles) are under-trained under uniform sampling because their episodes are short, so they contribute few transitions to the replay buffer.

2.4 Strategy 3: Adaptive sampling + stability-shaped reward

Same sampler as Strategy 2, but the reward is $r_t = 1 - 0.5\,\frac{|\theta_t|}{\theta_{\max}} - 0.25\,\frac{|x_t|}{x_{\max}}$ while alive and $r = -1$ on failure, with $\theta_{\max} = 12^\circ$ and $x_{\max} = 2.4$ the termination limits. The shaping prefers an upright, centred pole over a pole that merely has not fallen yet. Our hypothesis is that this margin is what lets the policy absorb the wind bumps and the increased force of the test phases, in the spirit of robust RL [6], without changing any training dynamics. Note that this shaping is not potential-based [5], so it deliberately changes the optimal policy.

2.5 Experimental setup

Each strategy is trained with 3 independent seeds (0, 1, 2) for 80,000 environment steps. Every selected network is evaluated with the unmodified course test protocol: 30 lengths in $[0.4, 1.8]$, 10 episodes each, wind phase from step 500, increasing force from step 1000, episode runs until failure. As an additional baseline we evaluate an untrained (randomly initialised) network. We report the mean over lengths, the per-length mean ± std, and test differences between strategies with a two-sided Wilcoxon signed-rank test over the 30 seed-averaged per-length means (paired by length) and a Mann–Whitney U test over all $30 \times 10 \times 3$ episodes. Code and the trained networks are available at https://github.com/Danila1801/REPO.

3 RESULTS AND DISCUSSION

Table 1 and Figure 1 summarise the test results; the short / mid / long bands are $\ell \in [0.40, 0.86]$, $[0.88, 1.33]$ and $[1.37, 1.80]$. All trained agents are far above the untrained network (20 steps; $p < 10^{-8}$ for every comparison), so all three strategies learn a transferable controller. Averaged over seeds, **adaptive sampling with the stability-shaped reward performs best** (10,065), followed by uniform domain randomisation (6,531) and adaptive sampling with the plain reward (2,246). Over the 30 lengths the shaped agent beats uniform on almost every length (Wilcoxon $W = 6$, $p = 2.6 \cdot 10^{-8}$) and adaptive on every length ($W = 0$, $p = 1.9 \cdot 10^{-9}$); uniform beats adaptive ($W = 2$, $p = 5.6 \cdot 10^{-9}$). The Mann–Whitney tests over all 900 episodes per strategy agree (all $p < 10^{-12}$).

Table 1: Test score (mean episode length over 30 lengths × 10 episodes). Seed mean ± std over 3 seeds; best seed.

Strategy	Seed mean	Best seed	Short / Mid / Long ℓ
Untrained network	20	–	19 / 21 / 19
Uniform (baseline)	6,531 ± 7,252	16,745	2,442 / 6,618 / 10,533
Adaptive	2,246 ± 1,343	3,222	1,359 / 3,017 / 2,363
Adaptive + shaped	**10,065** ± 3,785	14,729	6,459 / 10,957 / 12,777

Stability matters more than where we practise. The largest gain comes from the reward, not the sampler. All agents are trained only on 500-step episodes with default force, yet the test is decided after step 500, when the force magnitude jumps to 75 and later grows further. A +1 reward is indifferent between a pole that wobbles near ±12° and one held upright, so such agents often reach 500 steps with little margin and fall at the first strong push. The shaped reward makes the centred, upright state strictly better, which is exactly the margin the perturbation phases require. This is also visible in the validation statistic: shaped agents had a mean $|\theta|$ of 0.003–0.012 rad at their selected checkpoints versus 0.041–0.083 rad for the other two strategies. Notably, the one strong uniform seed (16,745) was also the steadiest uniform agent (0.043 rad), consistent with the view that a steady pole, not survival alone, predicts robustness. The gain is largest on short poles (6,459 vs. 2,442 for uniform), where the dynamics are fastest and a small margin is decisive.

Regret-based sampling did not help, and hurt. Contrary to our hypothesis, adaptive sampling with the plain reward was the weakest strategy. We see two reasons. First, the competence signal saturates: once the agent survives 500 steps on a length, $c_k \to 1$ for that bin and the sampler becomes almost uniform again (final p_k ranged only between 0.05 and 0.08), so the curriculum cannot tell a fragile controller from a robust one. Second, in early training it over-samples short poles, whose episodes end quickly; the replay buffer is then dominated by short failure episodes, and with the plain reward the Q-targets give little information about long-horizon balance. With the shaped reward the same sampler works well, which suggests the reward, not the sampling, carries the useful signal.

Variance across seeds. The best single network was a uniform agent (16,745), but the other two uniform seeds reached only 614 and 2,234. With only 3 seeds the seed-level difference between shaped and uniform is not significant (Welch t-test $p = 0.58$; shaped vs. adaptive $p = 0.09$), so the per-length and per-episode tests above should be read as evidence about these trained networks, not as a definitive ranking of training procedures. The shaped strategy, however, is more *reliable*: its worst seed (5,457) is far better than the worst uniform (614) and adaptive (347) seeds. DQN with a +1 reward is known to be unstable on CartPole as Q-values grow towards $1/(1-\gamma)$; the bounded, informative shaped reward seems to reduce this. Across all strategies, short poles remain hardest and the standard deviation per length is large, because a single unlucky wind push ends an episode.

Limitations and future work. We used 3 seeds and one training budget; more seeds would allow a proper seed-level test. Natural next steps are a competence signal that does not saturate (e.g. TD-error, as in PLR [3]), episodes longer than 500 steps during training, and an ablation of the shaping weights. In conclusion, to build a robust CartPole generalist with a fixed small network, rewarding *how well* the pole is balanced mattered much more than choosing *which* pole lengths to practise on.

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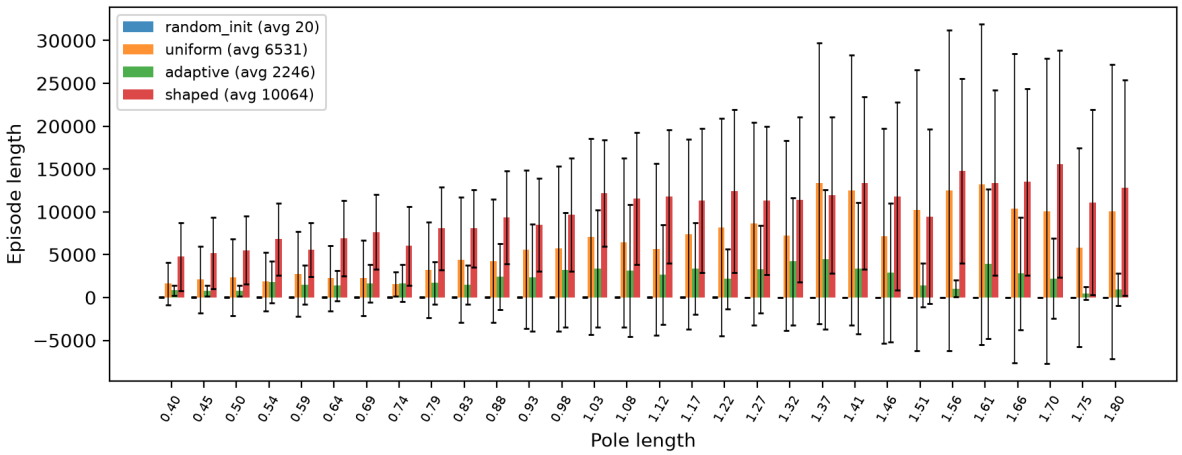


Figure 1: Per-length test score (mean ± std over 3 seeds × 10 episodes) for all strategies.

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A BAR PLOTS PER STRATEGY (BEST SEED)

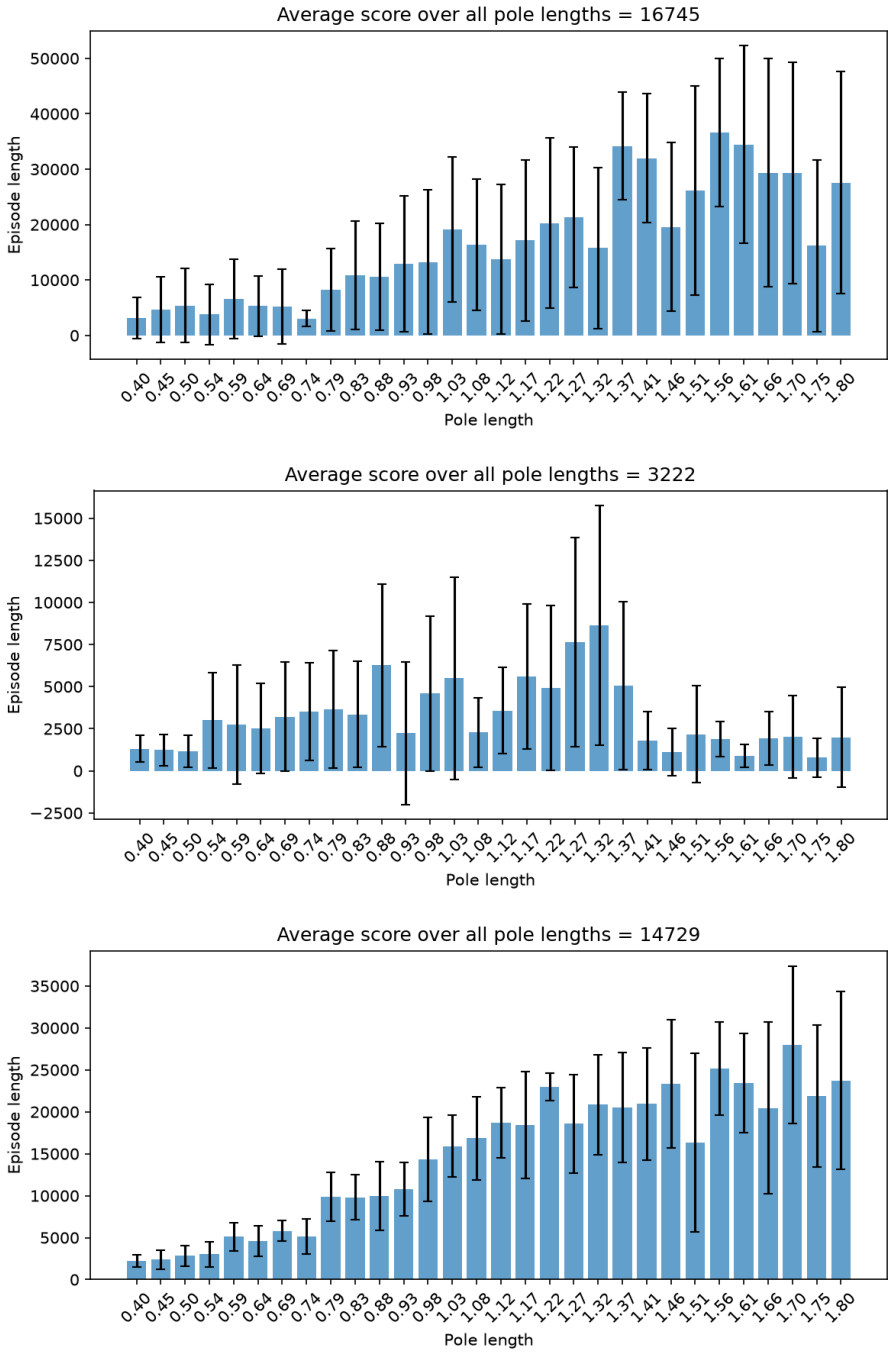


Figure 2: Course-format bar plots (avg ± std over 10 episodes) of the best seed per strategy. Top: uniform (seed 2), middle: adaptive (seed 1), bottom: adaptive + shaped (seed 2).